PHYSICAL SHORT-DISTANCE ASYMPTOTICS OF THE LATTICE YANG–MILLS ACTION DENSITY

Abstract. We derive a two-point action-density comparison at a prescribed short-distance scale from explicit local comparison and running coefficient estimates. The source factor is the derivative of the actual running coupling shift with the real reference data fixed. The normalized free curvature covariance gives the coefficient $9/\pi^4$ for $SU(2)$. Exact source differentiation, local response estimates and a conditional reference-law comparison control the physical errors relative to the leading fourth power of the running coupling. A common holomorphic cell normalizer transfers the comparison to every admitted periodic Gibbs state at the same bare datum. The result gives a logarithmically decreasing ultraviolet upper bound and a positive lower bound for separated nonnegative tests. The actual local conditional law, history estimates and running coefficient bounds remain explicit hypotheses; continuum existence is not asserted.

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1. Introduction

The short-distance covariance of the Wilson action density has two distinct sources of normalization. The quadratic curvature law fixes its free coefficient. The response of the extracted couplings to a scalar source fixes the factor multiplying each physical insertion. Both must be identified before a comparison with a Gaussian measure has a definite meaning. A bound that is merely small in absolute terms is insufficient when the leading covariance itself tends to zero with the running coupling.

We first fix the physical observable and its free normalization. The running coefficient estimates then select the observation depth and determine the constant-source response with all real reference data held fixed. Differentiation of the actual retained source weight separates the covariance of its first logarithmic jets from its mixed logarithmic jet. The local comparison hypotheses control these terms, including the contributions of retained histories and the conditional exterior field.

The passage from a reference torus to arbitrary periodic Gibbs states is a separate step. A common cell normalizer permits source derivatives to be compared without estimating their possibly divergent individual means. Signed copies of a test pair improve the resulting state difference by the number of disjoint copies. The error is then negligible relative to the positive free signal on the full prescribed shape class.

The extraction of a scalar curl coefficient from an effective action follows the polarization analysis of Bałaban [6, §5, pp. 292–297]. That work treats a small-field approximation. It does not provide

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the physical conditional-law and history estimates assumed here. We use the free curvature calculation of [3], the admitted trajectory and normalization conventions of [4], and the fixed-source and state-transfer results of [5], with reflection and chessboard estimates from [1]. The hypotheses below state the additional local comparison and coefficient information needed for the physical law. Their establishment and continuum reconstruction are separate problems.

2. Physical observables and fixed source data

Fix an odd integer block factor $L \geq 3$. At cutoff depth $K \in \mathbb{N}$, the mesh is $a = L^{-K}$. A compatible torus is $\Lambda_{a,\ell} = (a\mathbb{Z}/\ell\mathbb{Z})^4$, with ℓ/a integral and with the block divisibilities required at the depths in use. One independent bond variable lies in $SU(2)$ for each positively oriented nearest-neighbour bond; reversing a bond replaces its variable by its inverse. Haar measure has mass one. For a positively oriented plaquette p, let U_p be its ordered bond product and x_p its barycentre. Put

$$A(U) = \sum_p a_p(U), \qquad a_p(U) = 1 - \tfrac{1}{2}\operatorname{tr} U_p, \qquad \mathcal{O}_a(f) = \sum_p f(x_p) a_p(U). \tag{2.i}$$

Here f is a compactly supported real Lipschitz function, or its periodic version on the torus. The continuum test class will be the C^1_c subclass. The six positive plaquette orientations are all included. The trace is real on $SU(2)$, and $0 \leq a_p \leq 2$. In particular, every fixed-cutoff compact insertion is bounded.

The torus Wilson measure is the normalized measure $e^{-\beta A(U)} dU$, where $\beta \geq 0$. On the infinite lattice, an admitted state is a periodic Wilson Gibbs state at this same pair (a, β). Periodicity means invariance under a finite-index subgroup of bare lattice translations. The reflection theorem of [1, Theorem B] gives invariance under all bare lattice translations for these states. It does not identify every such state with a particular sequence of torus measures. For any of these probability measures ρ, write

$$S^T_{2;a,\rho}(f,h) = \rho(\mathcal{O}_a(f)\mathcal{O}_a(h)) - \rho(\mathcal{O}_a(f))\rho(\mathcal{O}_a(h)). \tag{2.ii}$$

This is a bilinear covariance, without complex conjugation. Complex sources are used to differentiate it; the comparison and positivity conclusions concern real nonnegative tests.

There is no additional a^4 in eq. 2.i. To check the convention, use traceless Hermitian colour coordinates with $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr}(XY)$. For a smooth field, $U_p = \exp(ia^2 F_{\mu\nu}(x_p) + O(a^3))$ implies $a_p = \frac{1}{2}a^4 |F_{\mu\nu}(x_p)|^2 + O(a^5)$. The plaquette sum therefore approximates $\frac{1}{2}\int f \sum_{\mu<\nu} |F_{\mu\nu}|^2$. This expansion fixes the normalization of the free quadratic density. It is not a smoothness assertion about the random field.

For finitely many tests, let $W_z = \sum_i z_i f_i$. On a finite torus set

$$Z_{a,\ell}(W_z) = \int \exp\{-\beta A(U) + \mathcal{O}_a(W_z)\}\, dU.$$

For an infinite-volume state use $\rho(e^{\mathcal{O}_a(W_z)})$ instead of the partition ratio. Both functions are entire in the finite source vector at fixed cutoff: the exponent is a finite linear combination of bounded observables, so its power series is uniformly dominated on every compact source set. Both have value one after normalization at zero, and hence have a local holomorphic logarithm there.

All real reference couplings, test cutoffs, averaging kernels, coordinate choices and history labels are fixed before the source is varied. In an exact blocked representation, its positive reference measure is likewise the unsourced measure. The source dependence is carried by the actual retained weight. The transport identity of [5, Proposition 3.1] supplies this representation when its fixed-kernel hypotheses hold. A new reference construction at each source value would require the derivatives of that construction as well and is not used here.

The running inverse coupling at a witness depth will be denoted by $x = x_K(d)$, with $g_K(d) = x^{-1/2}$. Its constant-source response is determined from the actual coefficient maps below. The

fixed small terminal coupling that selects the bare sequence is a different parameter. Constants may depend on that terminal choice, L, the fixed construction parameters and the fixed shape threshold. They are independent of cutoff depth, admitted state, test position and test amplitudes unless a statement says otherwise. Uniform error functions are always stated on the admitted parameter range. All logarithms are natural.

3. The separated free comparison

3.1. **Test geometry.** Write $T_\ell = \mathbb{R}^4/\ell\mathbb{Z}^4$, with its flat distance and unnormalized volume measure. Let $Y = \mathbb{R}^4$ or T_ℓ.

Definition 3.1 (Separated test pairs)**.** For $d > 0$ and $0 < \vartheta \le 1$, let $\mathfrak{T}_Y(d;\vartheta)$ consist of pairs (f,h) of nonzero, nonnegative Lipschitz functions. Each support lies in a ball of radius d. Write $M_v = \|v\|_\infty$ and $m_v = \int_Y v$ for $v = f, h$. The requirements are

$$\begin{aligned} d \le \operatorname{dist}(\operatorname{supp} f, \operatorname{supp} h) \le 6d, \\ m_v \ge \vartheta M_v d^4, \qquad \operatorname{Lip}(v) \le M_v/(\vartheta d). \end{aligned} \tag{3.i}$$

On T_ℓ take $\ell \ge 40d$. The C^1_c class used in the continuum application is contained in this class with the same parameter $\vartheta_* = 1/12000$. ■

Lemma 3.2 (Size and scaling of the comparison integral)**.** For $(f,h) \in \mathfrak{T}_Y(d;\vartheta)$, the two supports lie in a common ball of radius $5d$. On their product the distance lies in $[d, 10d]$. With $v_4 = \pi^2/2$, the integral

$$\mathcal{K}(f,h) = \iint f(x)h(y)|x-y|^{-8}\,dx\,dy$$

satisfies

$$10^{-8}\vartheta^2 M_f M_h \le (10d)^{-8} m_f m_h \le \mathcal{K}(f,h) \le d^{-8} m_f m_h \le v_4^2 M_f M_h. \tag{3.ii}$$

On a torus the integral is evaluated using a common lift of the supports to $\mathbb{R}^4$. It is independent of the lift. For $t > 0$ let $v_t(x) = v(x/t)$ on tY. Then $(f_t, h_t) \in \mathfrak{T}_{tY}(td;\vartheta)$ and $\mathcal{K}(f_t,h_t) = \mathcal{K}(f,h)$. ■

Proof. Choose points attaining the minimum cross-support distance. The distance of the two supporting ball centres is at most $d + 6d + d = 8d$. Choose lifts of those centres realizing that distance when Y is a torus. Their midpoint has distance at most $5d$ from both lifted balls. Any difference in this larger ball has length at most $10d < \ell/2$. Its Euclidean length therefore equals the flat torus distance. This identifies the integral with the one using the flat distance intrinsically. The lower separation follows from eq. 3.i and the upper separation from the diameter of the common ball. Integration against the nonnegative product gives the two middle inequalities. The mass floor and the volume of each support ball give the outer inequalities. Under dilation the distances are multiplied by t, the Lipschitz constants by t^{-1} and the masses by t^4. The factor t^8 in the product measure cancels the homogeneity t^{-8} of the kernel. □

3.2. **Curvature normalization.** In the pairing of Section 2, take an orthonormal basis of Hermitian traceless colour generators T_α, $1 \le \alpha \le 3$, with $\frac{1}{2}\operatorname{tr}(T_\alpha T_\beta) = \delta_{\alpha\beta}$. For a raw link increment A, the leading plaquette logarithm of e^{itA} is $it\,\mathrm{d}A$, where d is the unscaled oriented link circulation. Its trace vanishes at every order. Expanding the exponential in the Wilson action gives

$$1 - \tfrac{1}{2}\operatorname{Re}\operatorname{tr} U_p(t) = \tfrac{1}{2}t^2|(\mathrm{d}A)(p)|^2 + O(t^3).$$

Thus the free Wilson curvature at coupling g is the three-colour Gaussian with covariance $g^2 P_{\mathrm{ex}}$ per colour, where P_{ex} is the counting-coordinate orthogonal projection onto $\operatorname{ran}\mathrm{d}$. Indeed d is an isomorphism on $(\ker \mathrm{d})^\perp$ onto this range, and its constant Jacobian cancels in the normalized Gaussian integral.

For one colour, differentiating $\delta_{\mu\nu}/(4\pi^2|u|^2)$ once in each curvature argument gives the unit-coupling continuum curvature matrix on $\Lambda^2\mathbb{R}^4$. In detail, with $G(u) = (4\pi^2|u|^2)^{-1}$ and $J = \widehat{u}\widehat{u}^{\mathsf{T}}$,

its action is the negative of $\nabla^2 G = (4J - I)/(2\pi^2|u|^4)$ on each factor of the wedge, added together. Since J has rank one, the term $Jv \wedge Jw$ vanishes. Consequently

$$\mathsf{K}(u) = \frac{\Lambda^2(I - 2J)}{\pi^2|u|^4}, \qquad \|\mathsf{K}(u)\|_{\mathrm{HS}}^2 = \frac{6}{\pi^4|u|^8}, \qquad C_G = \frac{9}{\pi^4}. \tag{3.iii}$$

The norm calculation uses orthogonality of $I - 2J$ and the six positive plane labels. For real centred Gaussian coordinates X, Y of any covariance, $\mathrm{Cov}(X^2, Y^2) = 2(\mathbb{E}XY)^2$. The two half-density factors give $1/4$, and summing over independent colours gives 3. Thus the separated half-square covariance at coupling g is $C_G g^4 \mathcal{K}(f, h)$. This calculation concerns only separated correlations.

Proposition 3.3 (Free reference at a blocked scale)**.** On the periodic fine lattice of spacing $a = L^{-K}$ and side ℓ, use a compatible block hierarchy of odd factor L. Take $0 \le j \le K$, $b = L^j a$, and for $j \ge 1$ require at least two sites per direction at mesh b. Let P_j be the unit-coupling counting-coordinate covariance of the conditional mean of the exact free curvature given its level-j plaquette sheet averages. In particular $P_0 = P_{\mathrm{ex}}$. Use the barycentre x_p of each positively oriented fine plaquette p. Define the free reference sum by

$$\mathcal{K}_{a,j,\ell}(f, h) = \sum_{p,q} f(x_p) h(x_q) P_j(p, q)^2.$$

Then the connected covariance of the centred free half-squares of that conditional mean at coupling $g > 0$ is exactly $\frac{3}{2} g^4 \mathcal{K}_{a,j,\ell}(f, h)$. For $(f, h) \in \mathfrak{T}_{T_\ell}(d; \vartheta)$ with $10a \le d \le \ell/100$ and $b \le d$, it satisfies

$$\left| \frac{(3/2)\mathcal{K}_{a,j,\ell}(f, h)}{C_G \mathcal{K}(f, h)} - 1 \right| \le C_L \left[(a/d)^{1/2} + (d/\ell)^4 + \frac{a}{\vartheta^3 d} + \mathbf{1}_{\{j \ge 1\}} e^{-c_L d/b} \right]. \tag{3.iv}$$

The constants are uniform in a, j, ℓ, g, positions and amplitudes. ■

Proof. The orthogonal splitting for the actual sheet averages in [3, Proposition 3.2] identifies P_j. The Gaussian square identity gives the exact covariance above. The counting entries $P_j(p, q)$ become physical entries after multiplication by a^{-4}, since the physical curvature is a^{-2} times the raw flux. This change supplies a^8 in the double quadrature for the squared physical entries. In these conventions [3, Theorem D] gives eq. 3.iv. In its application here, every cross-displacement lies in $[d, 10d]$ by Lemma 3.2. Thus the pointwise comparison range $[10a, \ell/10]$ contains every contributing pair, including pairs whose support balls meet a seam of the torus. The lost fluctuation kernel has the exponential bound in that theorem at distance $r \ge b$. The shifted-grid quadrature has absolute cost $CaM_f M_h/(\vartheta d)$. Dividing by eq. 3.ii gives the relative cost $Ca/(\vartheta^3 d)$. □

Lemma 3.4 (Bounds for the raw background and slice kernels)**.** Use the hierarchy and the counting-coordinate projections of Proposition 3.3. At level j put $b_j = L^j a$, and let r be the flat torus distance between the fine plaquette barycentres. Uniformly over the compatible tori and $0 \le j \le K$,

$$\begin{aligned} &|P_j(p, q)| \le C_L (a/b_j)^4, \\ &|P_j(p, q)| \le C_L a^4 r^{-4} \qquad (\max(10a, b_j) \le r \le \ell/10). \end{aligned} \tag{3.v}$$

For $j \ge 1$ the slice satisfies

$$|(P_{j-1} - P_j)(p, q)| \le C_L (a/b_{j-1})^4 e^{-c_L r/b_{j-1}}.$$

■

Proof. Write $n = L^j = b_j/a$. In raw counting coordinates let $H_n B$ be the least-norm fine exact curvature with sheet datum $\mathrm{d}_c B$, where d_c is the coarse circulation. If the fine torus has just one site, all raw circulations vanish, so we may work with at least two sites at every used level. The range of H_n is $\operatorname{ran} P_j$ by the sheet identity and the least-norm characterization. In fact the image

of the sheet map on exact fine curvatures is the range of d_c, since the link average is onto. The lift bound proved in [2, proof of Proposition 6.4] gives

$$\sum_e |H_n(p,e)| \le C_L n^{-2}.$$

Here e ranges over coarse bonds. The bound follows by summing the kernel $C_L n^{-2} e^{-c_L \operatorname{dist}(p,e)/b_j}$ on the coarse lattice. The sum of its periodic images is bounded by the sum on the full coarse lattice. The lift estimate is at mesh b_j and uses no step above that mesh. For $n = 1$ the lift is the ordinary plaquette circulation and the row bound follows by counting its four entries.

Set $\mathcal{E}_n = H_n^{\mathsf{T}} H_n$ and use $^+$ for the inverse on $(\ker \mathrm{d}_c)^\perp$, extended by zero on $\ker \mathrm{d}_c$. The kernel is the same for $\mathcal{E}_n$ and d_c by the defining datum and least-norm property. By [2, Proposition 4.1], $\mathcal{E}_n \ge \mathrm{d}_c^{\mathsf{T}} \mathrm{d}_c$. Order reversal on the quotient gives $\mathcal{E}_n^+ \le (\mathrm{d}_c^{\mathsf{T}} \mathrm{d}_c)^+$. If there are $N = \ell/b_j \ge 2$ sites per direction, the diagonal entries of the latter are bounded by

$$\frac{1}{N^4} \sum_{k \ne 0} \frac{1}{4 \sum_{\mu=1}^4 \sin^2(\pi k_\mu / N)} \le \frac{C}{N^2} \sum_{0 < |k|_\infty \le N/2} |k|^{-2} \le C.$$

The frequencies are represented with $|k_\mu| \le N/2$. The eigenvalue displayed is the positive curl eigenvalue on the Fourier complement of the discrete gradients, and the zero frequency is removed. The last estimate counts at most Cm^3 frequencies with $|k|_\infty = m$. Positivity and Cauchy–Schwarz then bound every entry of $\mathcal{E}_n^+$ by C. Apply the row bound twice to $P_j = H_n \mathcal{E}_n^+ H_n^{\mathsf{T}}$. This proves the first line of eq. 3.v. All matrices here use raw flux and counting pairings.

The slice bound is [2, Proposition 6.4], expressed in the distance r by moving the plaquette base points by $O(a)$. At the distances of the second line, [3, Proposition 8.3] and eq. 3.iii give $|P_0(p,q)| \le C a^4 r^{-4}$. For $1 \le i \le j$, put $t_i = r/b_{i-1}$. The sum of the slice bounds is at most $C_L a^4 r^{-4} \sum_i t_i^4 e^{-c_L t_i}$. Since $r \ge b_j$, all $t_i \ge 1$. Use $t_i^4 e^{-c_L t_i} \le C_L / t_i$ and the geometric sum of b_{i-1}/r to bound the last sum uniformly. Subtracting the slices from P_0 proves the second line. □

4. The witness scale and the source coefficient

Fix a terminal floor $H > 1$ and a first-passage trajectory $x_0 = \beta, \dots, x_K$ of the real construction in [4, Theorem A]. Use the same formed coefficients in every admitted finite realization of that prefix. Thus $H < x_K \le H + B$, and there are fixed $\lambda, B > 0$ and $C_0 \ge 0$ such that

$$\lambda(j-i) - C_0 \le x_i - x_j \le B(j-i) \qquad (0 \le i \le j \le K). \tag{4.i}$$

The actual increments have a common one-loop part b_i^0, whose partial sums differ from ic_0 by at most a fixed constant, and a remaining part bounded at step i by $e(x_{i-1})$, where the nonnegative envelope e is bounded on $[H, \infty)$ and tends to zero at infinity. In the half-trace Wilson convention,

$$c_0 = \frac{11}{3\pi^2} \log L.$$

These are the real trajectory and coefficient inputs of [4]. The one-loop coefficient is a determinant and trace input in that construction.

Write $\eta_j = L^{j-K}$ for the physical mesh at level j, and let M be the fixed side of a localization cube in its own mesh units. Fix the shape parameter $\vartheta > 0$ and the positive logarithmic exponent p of the small-field radius. Define

$$\overline{x}_s = H + B(s+1), \qquad D(s) = D_0 (\log \overline{x}_s)^{2p}, \qquad s(d) = \min\{s \in \mathbb{Z}_{\ge 0} : L^s d \ge D(s)\}. \tag{4.ii}$$

The fixed constant D_0 is chosen after L, M, ϑ, with room for the localization and source ranges specified below. Choose it so that $D(s) \ge 1$. This can be done uniformly for $H \ge e$.

Proposition 4.1 (Choice of scale)**.** For $0 < d < 1$, let $s = s(d)$ and use a cutoff with $K \geq s$. Set

$$k_0 = K - s, \qquad a_s = L^{-s}, \qquad \widehat{d} = d/a_s, \qquad x = x_{k_0}, \qquad g_K(d) = x^{-1/2}.$$

Then $D(s) \leq \widehat{d} < LD(s)$, and at fixed H,

$$\frac{s(d)\log L}{\log(1/d)} \longrightarrow 1, \qquad \frac{g_K(d)^{-2}}{\log(1/d)} \longrightarrow c_* := \frac{11}{3\pi^2} \qquad (d \downarrow 0), \tag{4.iii}$$

uniformly over the admitted first-passage data with $K \geq s(d)$. In particular $\widehat{d}$ is bounded above and below by fixed positive multiples of $(\log x)^{2p}$ for small d. If

$$m = \lceil 2\log_L \overline{x}_s \rceil, \qquad k_1 = k_0 + m,$$

then $k_1 \leq K$ for small d, and $\sup_{k_0 \leq j \leq k_1} |x_j/x - 1| \to 0$ uniformly in the same data.

Suppose the complete clean reads at these levels have a joint lift of diameter at most $P(s)L^{-(s-m)}$, where P is a fixed polynomial with nonnegative coefficients. On any fixed reference torus they fit in a common injective patch for all small d. For a shape test f, its ordinary source norm at level j is

$$\|f\|_{\sharp, K-j} = \max(\|f\|_\infty, c_M \eta_j \operatorname{Lip} f),$$

where c_M is the fixed locality constant of [5]. For D_0 sufficiently large this is $\|f\|_\infty$ at $j = k_0$, and is at most $L^{j-k_0}\|f\|_\infty$ for $k_0 \leq j \leq k_1$. ∎

Proof. Since $\log D(s) = O_H(\log\log(s+2))$, the exponential $L^s d$ eventually exceeds $D(s)$. The minimum is positive when $d < 1$. Its defining inequality and the failure at $s-1$ give the two bounds on $\widehat{d}$, using that D is nondecreasing. They give $s\log L = \log(1/d) + O_H(\log\log(s+2)+1)$. Also $s \to \infty$ as $d \downarrow 0$, since each fixed set of candidate levels fails for small enough d. The first limit follows on dividing by $s\log L$.

For the second limit, write $e_*(y) = \sup_{t \geq y} e(t)$ on the admitted half-line. It is bounded by hypothesis. By eq. 4.i, $x_{K-u} \geq H + \max(0, \lambda u - C_0)$. Telescoping the last s increments yields

$$|x_{K-s} - x_K - c_0 s| \leq C + \sum_{u=1}^{s} e_*(H + \max(0, \lambda u - C_0)) = o(s).$$

The last bound follows by averaging a bounded sequence tending to zero, independent of the datum. The bounded terminal interval and the first limit give eq. 4.iii. The lower drift and the upper envelope $\overline{x}_s$ give $x \asymp_H 1 + s$ and $\overline{x}_s \asymp_H x$. This gives the stated size of $\widehat{d}$.

At fixed H, $m = O_H(\log(s+2))$ and $L^m \leq L\overline{x}_s^2$. The drift bounds give $|x_j - x| \leq Bm + C_0 = o(x)$ through k_1. The lifted diameter tends to zero since $P(s)L^{m-s} \leq LP(s)\overline{x}_s^2 L^{-s}$. When it is less than a quarter of the reference period, a ball about a point of the read contains the lift and projects injectively. This is a geometric consequence of the supplied bound on the complete reads. Finally, the shape condition gives $c_M\eta_{k_0}\operatorname{Lip} f \leq c_M\|f\|_\infty/(\vartheta\widehat{d})$. Choose D_0 so the last prefactor is at most one. The bound at $j \geq k_0$ then follows from $\eta_j = L^{j-k_0}\eta_{k_0}$. □

Corollary 4.2 (One-loop comparison and dilation)**.** At fixed H, let

$$g_{\mathrm{lin}}(d) = \big(H + c_* \log(1/d)\big)^{-1/2}.$$

Uniformly on the first-passage data of proposition 4.1, $g_K(d)/g_{\mathrm{lin}}(d) \to 1$. In a physical comparison with main factor $g_K(d)^4$, this replacement multiplies the existing relative factor by $(g_K(d)/g_{\mathrm{lin}}(d))^4$.

Let K' be the first-passage depth for the same bare coupling β, and let $b = K - K'$. The Wilson measure on the torus of side ℓ at cutoff K is carried by lattice dilation to the torus of side $L^b\ell$ at cutoff K'. For each test put $f^{[b]}(x) = f(L^{-b}x)$. Then

$$S^T_{2;K,\ell,\beta}(f,h) = S^T_{2;K',L^b\ell,\beta}(f^{[b]}, h^{[b]}).$$

The shape parameter and the free integral $\mathcal{K}$ are preserved, with observation scale $L^b d$ on the right. A first-passage comparison therefore applies on the right when the dilated datum satisfies its volume, scale and cutoff hypotheses. ■□

Proof. By eq. 4.iii, the two inverse squared couplings divided by $\log(1/d)$ have the same positive limit. This gives the ratio and the stated relative factor.

Dilation by L^b maps spacing L^{-K} to $L^{-K'}$ and bijects oriented links and plaquettes. It preserves each plaquette product and each Haar factor. The normalized measures are therefore identified at the same β. The transformed test on each corresponding barycentre gives the same observable. The support radii and distance multiply by L^b, each mass by L^{4b}, and each Lipschitz constant by L^{-b}. In the free integral the two volume factors cancel the scaling of $|x-y|^{-8}$. These equalities hold also for negative b. □

4.1. **A constant source in the fixed reference.** Freeze the reference data of the unsourced trajectory at β, including $\theta_i = x_i$, the cubes, the cuts, and the readout maps. On a constant interior read, let $b_i(v)$ be the actual Wilson coefficient of the complete formed member at step i when the initial scalar source is v. The coefficient is read at its own pointing before its Wilson subtraction. The formation includes the marked contribution from earlier retained remainders. The coefficients are those of the common companion, so the interior scalar values agree on the admitted volumes. In particular $b_i(0) = x_{i-1} - x_i$.

We use the fixed-reference analytic and real matching hypotheses of [5, Propositions 4.1 and 4.2]. For the analytic part, the ordinary coefficient is the same formed holomorphic map of the old weight deviations in the sourced and unsourced presentations. More precisely, it has the form

$$b_i(v) = b_i^0 + \mathcal{B}_i((\theta_l - x_l + Z_l(v))_{0\le l<i}) + R_i(v),$$

where b_i^0 is source independent, Z_l is the shift defined in eq. 4.iv, and R_i is the marked part. The ordinary map $\mathcal{B}_i$ has no additional dependence on v. It is holomorphic on the full relative product of weight discs. The half product has room 6ϱ beyond the real base weights, for a fixed source radius $\varrho > 0$. On the half product its absolute derivative row sum is bounded by $C x_{i-1}^{-3/2}(\log x_{i-1})^p$. This is the regular coefficient bound on the actual creation domains. The marked coefficient is holomorphic and bounded by $C x_{i-1}^{-p_R}$ on $|v| < 4\varrho$, with $p_R > 1$. The coefficient maps preserve reality. For matching, the real weights have 2ϱ of room in the common tube and the full increments beyond the common one-loop part are bounded by $\varrho/3$. The shifted bare input is admitted and each subsequent weight in that tube has actual local successor admission. The two real presentations have common hereditary reads, centered formation, real cuts, vacuum convention and cofinal physical coefficient readout as in the same-frame source theorem of [5]. The numerical room does not supply these local admission tests.

Proposition 4.3 (The physical source factor)**.** Under these hypotheses, above a fixed floor for H, set

$$Z_j(v) = v + \sum_{i=1}^{j}(b_i(v) - b_i(0)), \qquad \zeta_j = Z_j'(0). \tag{4.iv}$$

The constant-source weight at level j is $x_j - Z_j(v)$. For the shifted real input, through the same fixed prefix,

$$x_j^{\mathrm{fix}}(\beta - v) = x_j - Z_j(v) \qquad (|v| < \varrho).$$

In particular

$$\zeta_j = \frac{d}{db} x_j^{\mathrm{fix}}(b)\Big|_{b=\beta} = 1 + \sum_{i=1}^{j} b_i'(0), \qquad \sum_{i=1}^{j} |b_i'(0)| \le C\big[x_j^{-1/2}(\log x_j)^p + \varrho^{-1} x_j^{1-p_R}\big]. \tag{4.v}$$

The floor can be chosen so that $2/3 \le \zeta_j \le 4/3$. At the witness set $\zeta_K(d) = \zeta_{k_0}$. Then $\zeta_K(d) \to 1$ as $d \downarrow 0$, uniformly over the admitted first-passage data with $K \ge s(d)$. If $p_R \ge 3/2$, its difference from one is bounded by $C g_K(d)(\log g_K(d)^{-2})^p$. ■

Proof. At the bare level the source changes the coefficient in the Wilson weight from β to $\beta - v$. The coefficient subtracted from the regular member is added to the exponent of that weight as its own pointed Wilson action. Thus the inverse weight loses $b_i(v)$ at step i. The stage sums of tents equal one on the interior read. The resulting weight is $\beta - v - \sum_{i \le j} b_i(v) = x_j - Z_j(v)$. For real v, apply the same-frame source theorem of [5] with the stated hereditary admission and common readout. This identifies that weight with the real shifted presentation while the reference data stay fixed. The derivative in eq. 4.v follows by the chain rule.

For the estimate put $d_i = \sup_{0<|v|<4\varrho} |b_i(v) - b_i(0)|/|v|$. Assume inductively that $\sum_{l<i} d_l \le 1/6$. Each previous weight shift then has modulus at most $(7/6)|v| < 6\varrho$. The segment between the unsourced and sourced old weights lies in the half product. Integrating its derivative row bound bounds the unmarked difference quotient by $C x_{i-1}^{-3/2} (\log x_{i-1})^p$. Subtracting the value at zero from the marked coefficient and applying Schwarz's lemma on $|v| < 4\varrho$ bounds its difference quotient by $C\varrho^{-1} x_{i-1}^{-p_R}$. The one-loop part is common and cancels.

For $i \le j$, the lower drift gives $x_{i-1} \ge x_j/2 + \lambda(j - i + 1)$ once $H \ge 2C_0$. Both envelope functions decrease above a fixed floor. For $A \ge e$ and $q > 1$, their required integral bound is

$$\int_A^\infty y^{-q} (\log y)^p \, dy = A^{1-q} \int_0^\infty e^{-(q-1)t} (\log A + t)^p \, dt \le C_{p,q} A^{1-q} (\log A)^p.$$

Comparison of the sums with this integral gives the right side of eq. 4.v as a bound on every radial sum. Increasing H makes this bound at most $1/6$ throughout the prefix. This closes the induction, starting with $Z_0(v) = v$. The derivative at zero satisfies $|b_i'(0)| \le d_i$. Reality yields the claimed interval for ζ_j. The witness limit follows from $x \to \infty$ in eq. 4.iii, and the last bound follows from $x^{1-p_R} \le x^{-1/2}$ when $p_R \ge 3/2$. □

4.2. **Differentiating the pointed classical part.** On an ordinary interior read at the witness, all the used stage profiles equal one. Here the action is evaluated on the background formed from the level-k_0 data, using its bare-mesh plaquettes. The witness mesh a_s controls the read range. For each step $j \le k_0$, let $h_y^{(j)}(p)$ be its nonnegative tent weights at the plaquette p of the classical read and let p_y be its fixed bare source anchor. The exact classical identification in [5] transports the pointed action to this read, with $\sum_y h_y^{(j)}(p) = 1$ and $\operatorname{dist}(x(p), x(p_y)) \le CM\eta_j$ when $h_y^{(j)}(p) \ne 0$. These statements apply on the complete common read, with its projected branch tests and contours. They concern the literal transported Wilson counterterms.

Proposition 4.4 (Pointed first derivatives)**.** For $W = z_1 f_1 + z_2 f_2$ with disjointly supported shape tests, define f_i^w on the bare-mesh plaquettes of the witness background by

$$\zeta_{k_0} f_i^w(p) = f_i(x(p)) + \sum_{j=1}^{k_0} b_j'(0) \sum_y h_y^{(j)}(p) f_i(x(p_y)). \tag{4.vi}$$

The first derivative of the sourced classical exponent at zero is the Wilson action weighted by $\zeta_{k_0} f_i^w$. The mixed derivative of that classical exponent vanishes. The discrepancy and support obey

$$\begin{aligned} |f_i^w(p) - f_i(x(p))| &\le CMa_s \operatorname{Lip}(f_i) \sum_{j \le k_0} |b_j'(0)|, \\ \{x(p) : f_i^w(p) \ne 0\} &\subset (\operatorname{supp} f_i)^{+CMa_s}. \end{aligned} \tag{4.vii}$$

Here E^{+r} denotes the closed r-neighbourhood of E.

For a bilinear form on these plaquette weights, with the physical quadrature at the bare mesh $a = L^{-K}$, write

$$\mathcal{B}(u,v) = a^8 \sum_{p,q} u(p)v(q)\, B(p,q),$$

assume $|B(p,q)| \le Ad^{-8}$ between the two enlarged supports. When $CMa_s \le d/4$, replacing both sampled tests by the profiles f_i^w changes this form by at most

$$CA\frac{M}{\vartheta\widehat{d}}\left(\sum_{j\le k_0}|b_j'(0)|\right)\|f_1\|_\infty\|f_2\|_\infty. \tag{4.viii}$$

The profile can have either sign. ■

Proof. At the plaquette p the source changes the inverse weight by minus

$$W(x(p)) + \sum_{j\le k_0}\sum_y h_y^{(j)}(p)\big[b_j(W(x(p_y))) - b_j(0)\big].$$

The weights and anchor locations are fixed. This gives eq. 4.vi. Each summand depending on a single anchor has mixed derivative proportional to $f_1(x(p_y))f_2(x(p_y)) = 0$. The bare source is linear.

For the exact accounting, let $G_{j,y}(v)$ be the unnormalized pointed regular member on the fixed read and U its classical field. Its normalized exponent contains

$$G_{j,y}(v) - G_{j,y}(v)(\mathbf{1},0) - b_j(v)A(h_y^{(j)},U),$$

where $A(h,U)$ denotes the Wilson action with plaquette weight h and the exact inherited transport. The difference of the last term between $v = W(x(p_y))$ and $v = 0$ cancels precisely the contribution of the same coefficient to the shifted weight on the common read. Thus differentiating the bare source and the regular members before their Wilson subtractions gives the same answer as differentiating the running weight and the already subtracted members. In the latter representation, eq. 4.vi is the classical contribution. Its paired regular derivative is the derivative of the normalized member, with its own Wilson subtraction. Missing shares on other retained reads are separate terms in the complete exponent.

Subtract $\zeta_{k_0} f_i(x(p))$ on the right side of eq. 4.vi and use the unit sum of the tents. The difference at each anchor is bounded by $CM\eta_j \operatorname{Lip}(f_i)$. Since $\eta_j \le a_s$ and $\zeta_{k_0} \ge 2/3$, this proves the first inequality in eq. 4.vii. The range of the anchor read proves the second one.

The widened supports have separation at least $d/2$. Each contains at most $C(d/a)^4$ plaquette barycentres, with multiplicity over the six orientations. This bound follows by covering the enlarged ball by mesh cubes, with a bounded number of barycentres per cube. The first inequality in eq. 4.vii, the shape Lipschitz bound and the bound on the slope sum imply $\|f_i^w\|_\infty \le C_\vartheta\|f_i\|_\infty$ there. Expand the difference of products into $(f_1^w - f_1)f_2^w + f_1(f_2^w - f_2)$, with the original tests evaluated at the respective barycentres. In the estimate on each sum the factor $a^8 d^{-8}$ pays for both support counts. This yields eq. 4.viii. □

4.3. **Summing the ordinary first source derivatives.** In this subsection set $n = k_0$, $x = x_n$, and let f be one of the shape tests. Write $M_f = \|f\|_\infty$ and $M_i = \|f_i\|_\infty$ for their amplitudes. The physical meshes at levels $j \le n$ are $\eta_j = L^{j-K}$. Use the radii and the small-field profile of the actual source formation in [5], with

$$\delta_j = x_j^{-1/2}(\log x_j)^p, \qquad \alpha_j = x_j^{-1/2}(\log x_j)^q, \qquad s_j = \delta_j/\alpha_j.$$

Fixed radius constants can be absorbed into the estimates below. Take $q > 21p$ and $p > 1$. Formed regular and remainder origins have indices $1 \le j \le n$; their lists are empty when $n = 0$. On a support neighbourhood, assign each ordinary source term once in its own class and at its own root. A short ordinary read uses the same localized field and current as its whole-lattice formation at

that root. The complete determining set includes every branch test, chart read, gauge contour and source anchor. This common-read condition is required before replacing a history member by its formed jet. A local curvature test alone does not identify those two evaluations.

Lemma 4.5 (The summed ordinary source bounds)**.** Suppose the localized backgrounds at the witness are real and the passed test gives

$$|\log U_p| \le C(a/a_s)^2\delta_n$$

on the bare plaquettes of the localized read. Suppose the ordinary source formation and the hypotheses of [5, Proposition 5.6] hold on all the following reads.

(1) The near regular and defect terms are filed in witness windows meeting a fixed CMa_s enlargement of the support. In each window their evaluated complete field and current datum has size at most $C\delta_n$ and satisfies the affine comparison and truncation hypotheses of [5, Proposition 5.6]. At the older origins $\|f\|_{\sharp,K-j} \le M_f$ by the witness room.

(2) For an ordinary remainder of origin j, its actual source family is holomorphic on its radius $\varrho/\|f\|_{\sharp,K-j}$. Its bound there is $Cx_j^{-\kappa_0/2}e^{-\kappa l}$, where l is the length of its domain and $\kappa_0 > 4$. This is a bound both on its admitted evaluated pairs and on the product of that source disc with an old pair ball of radius $c\alpha_j$ about the trivial pair, on which the family is jointly holomorphic. The source family is invariant under constant colour rotations in that chart.

(3) A homogeneous nonwrapping read of age $A = n - j$ and length l has old pair norm at most $C(1+l)^b\delta_n L^{-2A}$ in its admitted axial frame. Its current is included in the pair norm. The ordinary long regular jets on these reads have the same admitted core and trivial-pair ball bounds as their weighted first-source amplitude. They vanish with their first field derivative at the trivial pair. Those outside the near windows have $l \ge cL^A - C$.

(4) At origin j the domain count at one root and length l is at most $Ce^{\kappa_D l}$, with $\kappa_D < \kappa$. The complete determining set together with its root has diameter at most $C(1+l)\eta_j$. For a nonhomogeneous ordinary read which reaches a distant live region or winds on the reference torus, require $l \ge c\widehat{d}L^A - C$. Use the unsubstituted source-disc bounds on those reads, with a fixed polynomial in x, L^A allowed in their individual majorants at rate κ. The ordinary two-source terms are holomorphic with these absolute majorants on a fixed source bidisc of radii proportional to M_i^{-1} and depend only on the source values in their declared complete determining sets. They include the terms in the class-independent scalar source energy before vacuum subtraction, as well as the remaining normalized interactions. The local scalar energy uses the same complete source reads and bounds, as in [5, Proposition 7.1].

All constants in these inputs are uniform in the cutoff and actual retained histories. Let $I(f)$ be the sum of the ordinary normalized first-source terms just specified, and $X(f)$ the localized Wilson read with profile from eq. 4.vi and its passed-test indicator. For any fixed b_0 with $0 < b_0 < \kappa_0/2 - 2$, the bounds are

$$\|X(f)\|_\infty \le Cx^{-1}\widehat{d}^4(\log x)^{2p}M_f,$$
$$\|I(f)\|_\infty \le Cx^{-1}\widehat{d}^4[(\log x)^{3p-q} + x^{-b_0}]M_f.$$

The ordinary mixed terms are bounded by $C\widehat{d}^4e^{-c'\widehat{d}}M_1M_2$. Terms filed as live, missing shares of the source number, and errors replacing the history background by its localization use separate bounds. ■

Proof. At a passed-test plaquette the half-trace action is bounded by $C(a/a_s)^4\delta_n^2$. The profile and support estimate eq. 4.vii bounds its weight by CM_f and the number of its bare plaquettes by $C(d/a)^4$. Multiplication gives the bound for $X(f)$.

The near-window bound in [5, Proposition 5.6] gives $C\delta_n^2 s_n M_f$ per witness window. It is used with the same source anchor for its complete pointed member and its own Wilson subtraction. At

most $C\widehat{d}^4$ such windows meet the enlarged support. The contribution of this part is therefore bounded by $C\widehat{d}^4 M_f x^{-1}(\log x)^{3p-q}$.

For the long regular terms, [5, Proposition 5.6] gives $A_j \le C\alpha_{j-1}^2 s_{j-1}$. Its pointed coefficient bound keeps the same norm after the member's own Wilson subtraction. The adjacent drift bounds give $\alpha_{j-1} \le C\alpha_j$, since $|x_{j-1} - x_j| \le B + C_0$. The lower drift gives $x_{j-1} \ge x_n - C_0$, so $s_{j-1} \le Cs_n$ since $q > p$. Using the older sharp norm bound in the statement, the regular jet majorant at these reads is thus $C\alpha_j^2 s_n M_f e^{-\kappa l}$.

For a homogeneous old domain, use its field and current ratio

$$u_j = C(1+l)^b(\delta_n/\alpha_j)L^{-2A}.$$

For an old remainder, source Cauchy and vacuum subtraction give the jet majorant

$$C\varrho^{-1}M_f x_j^{-\kappa_0/2}e^{-\kappa l}$$

on the old pair ball and on the evaluated pair. The subtraction kills the constant term. Its linear field term vanishes because the adjoint representation has no invariant linear functional. On the half ball of its old pair chart, Cauchy's formula bounds the evaluated jet by Cu_j^2 times its jet majorant. When u_j is greater than a fixed small threshold, the bound on the actual admitted core gives the same estimate after increasing the constant. This also applies to each homogeneous long regular jet, whose two vanishing terms are part of its normalized formation. No small field ratio is asserted on nonhomogeneous reads.

The radius-d support and the determining-set bound give at most $C(\widehat{d}L^A + 1 + l)^4$ possible roots at length l. Indeed a root of a read meeting the support is within distance $d + C(1+l)\eta_j$ of its centre, and $d/\eta_j = \widehat{d}L^A$. Use the bound $C\widehat{d}^4 L^{4A}(1+l)^4$. Summing u_j^2 times the remainder majorant over lengths with the strict entropy margin cancels L^{-4A} against L^{4A}. The resulting sum is at most

$$C\widehat{d}^4 M_f \delta_n^2 \sum_{j\le n} x_j^{-\kappa_0/2}\alpha_j^{-2}.$$

Since $\alpha_j^{-2} \le Cx_j$, comparison with an integral using $x_j \ge x/2 + \lambda(n-j)$ bounds the sum by $Cx^{2-\kappa_0/2}$. Its product with $(\log x)^{2p}$ is $O(x^{-b_0})$. The homogeneous long regular jets use $\alpha_j^2 s_n$ in place of the remainder factor. Their lower length bound leaves $e^{-c'L^A}$ after summing the length polynomial at the spare rate. Summing this in A gives $C\widehat{d}^4 M_f \delta_n^2 s_n$.

For a nonhomogeneous or winding read, sum the original majorant over its length lower bound. The resulting term has an $e^{-c'\widehat{d}L^A}$ factor. The root count and any fixed polynomial in $L^A, x, \widehat{d}$ are absorbed by a portion of this exponential, uniformly in age, because $\widehat{d}$ grows as $(\log x)^{2p}$. An ordinary term is independent of either source variable whose support misses its read. Thus the contributing mixed terms reach both supports and their lengths have the same lower bound. Cauchy's formula on the two source circles gives CM_1M_2 times the individual analytic bound. The same root and age summation proves the mixed estimate. The classical mixed terms vanish by proposition 4.4. □

5. The covariance on the reference torus

All the source derivatives in this section use the same bare datum and the same fixed auxiliary choices as in the unsourced run. We write $j_0 = k_0$, $j_1 = k_1$, $n = j_1 - j_0$, $x = x_{j_0}$, $g_x = x^{-1/2}$, and $\widehat{d} = d/a_s$ for the witness variables of proposition 4.1. Here $a_s = L^{-s}$ is the witness mesh and $a = L^{-K}$ is the bare mesh. Put $M_i = \|f_i\|_\infty$ for the two source profiles. A positive law on a retained state includes the standing class, the block field and all still stored slots. We keep these slots when differentiating source integrals.

Lemma 5.1 (Derivatives of the actual source integral)**.** At a stop j, suppose the actual summed source integral is

$$\frac{Z(z_1 f_1 + z_2 f_2)}{Z(0)} = \mathbb{E}_j \mathscr{W}_j(z), \qquad \mathscr{W}_j(0) = 1 \quad \text{almost surely},$$

where $\mathbb{E}_j$ uses the unsourced positive retained-state law. Assume $\mathscr{W}_j$ is measurable and pointwise holomorphic on a common source bidisc, and integrably dominated on every compact subdisc. Define its logarithmic jets at zero by

$$\Phi_i = \partial_i \mathscr{W}_j(0), \qquad \Phi_{12} = \partial_1 \partial_2 \mathscr{W}_j(0) - \Phi_1 \Phi_2.$$

Assume the products used below are integrable. Then the bare connected covariance satisfies

$$S_2^T(f_1, f_2) = \operatorname{Cov}_j(\Phi_1, \Phi_2) + \mathbb{E}_j \Phi_{12}. \tag{5.i}$$

If the actual retained weight is locally e^{Φ}, these are its first and mixed exponent derivatives. No common zero-free disc for all retained weights is required. ■

Proof. The compact-disc majorants and Cauchy's formula permit derivatives of $\mathscr{W}_j$ to pass under the integral. Its integral equals one at zero. The mixed derivative of its local logarithm is therefore

$$\mathbb{E}_j \partial_{12} \mathscr{W}_j(0) - \mathbb{E}_j \Phi_1 \, \mathbb{E}_j \Phi_2.$$

Substitute the definition of Φ_{12}. The bare integral gives the physical covariance by differentiating its bounded insertions. Each individual retained weight is nonzero on some neighbourhood of zero, so its local logarithmic derivatives agree with the displayed jets. That neighbourhood need not be uniform in the retained state. Integrating stored variables before computing these jets adds their conditional covariance to the mixed logarithmic derivative. The use of their actual retained source weight preserves this contribution. □

At the witness, write $\zeta = \zeta_K(d)$ and $\widetilde{f_i} = f_i^w$ for the profiles in eq. 4.vi. Proposition 4.4 gives their once-counted classical derivative and the vanishing of its mixed single-anchor terms.

Lemma 5.2 (Removal of the other witness source derivatives)**.** Use lemma 5.1 at j_0. Suppose the actual source filing gives

$$\Phi_i = c_i + \zeta X_i + I_i + B_i,$$

where c_i is the same scalar on all standing classes and X_i is the background Wilson read with weight $\widetilde{f_i}$ and its fixed local passed-test indicator. The scalar is the derivative of the complete scalar source energy. Its mixed derivative is included among the ordinary two-source terms of lemma 4.5. Missing shares on a retained class are included in B_i. Assume the complete ordinary read, response and summation hypotheses of lemma 4.5, with $p > 1$ and $q > 21p$. Suppose $\widehat{d} \le C(\log x)^{2p}$ and $|\zeta| \le 4/3$. Assume B_i and the nonordinary mixed derivative consist of tails bounded, after division by the same profile norms, by fixed powers of $\log x$ times $e^{-c\widehat{d}} + e^{-c\check{R}}$, and of terms supported on contact with a live component. On that event their respective pointwise majorants, after division by M_i and $M_1 M_2$, are

$$C(\log x)^C L^{C\check{R}}(1+A)^C L^{4A}, \qquad C(\log x)^C L^{C\check{R}}(1+A)^C L^{8A},$$

where A is the largest contacting lineage age. Suppose there are at most $C\widehat{d}^4$ possible contact roots and the actual age tails at each root are bounded by $Ce^{-c(\log x)^p}(2L^8)^{-a}$ for $A \ge a$. Assume $\check{R}/\log x \to \infty$, $\check{R} = o((\log x)^p)$ and $\widehat{d} \ge c(\log x)^{2p}$. Then, uniformly for these data,

$$S_2^T(f_1, f_2) = \zeta^2 \operatorname{Cov}_{j_0}(X_1, X_2) + o(x^{-2}) M_1 M_2. \tag{5.ii}$$

■

Proof. The ordinary mixed derivative has no pointwise classical part. The remaining ordinary mixed derivative is bounded by $C\widehat{d}^4 e^{-c\widehat{d}} M_1 M_2$ by lemma 4.5. The room makes it smaller than any inverse power of x. The same lemma gives the sup bounds for X_i and I_i, with some $b > 0$ in its inverse-power error.

On the contact event the age estimate gives

$$\mathbb{E}_{j_0}\big[1_{\text{contact}}(1+A)^C L^{8A}\big] \le C\widehat{d}^4 e^{-c(\log x)^p} \sum_{a\ge 0} (1+a)^C 2^{-a}.$$

We used the union bound over possible roots and the age tail at the same lineage whose fields appear in the pointwise majorant. This bounds $\|B_i\|_2/M_i$ and the mean nonordinary mixed derivative by $o(x^{-N})$ for every fixed N. The loss $L^{C\check{R}}$ is smaller than the rarity exponential, and both far exponentials are superpolynomial by the lower room and localization-radius bounds. Cauchy–Schwarz controls all covariance terms with a B_i. The constants c_i have zero covariance with every variable. Using $|\operatorname{Cov}(Y,Z)| \le 2\|Y\|_\infty \|Z\|_\infty$ for the terms with I_i leaves, after division by $x^{-2}M_1M_2$, at most

$$C\big[(\log x)^{21p-q} + (\log x)^{18p} x^{-b}\big].$$

Indeed the two support counts cost $(\log x)^{16p}$ and the background bound costs another $(\log x)^{2p}$. The square of the irrelevant bracket is absorbed by the bracket for large x. The right side tends to zero. Substitute these estimates in eq. 5.i. □

5.1. **Comparison of the clean retained law.** Let $u = a/a_s$. In the admitted small logarithm chart write the background plaquette on the retained datum at level j as $\exp(iu^2 Y_{j,p})$. Thus $Y_{j,p}$ is its traceless Hermitian curvature coordinate in units of u^2, with the half-trace norm of the physical setting. For $j = j_0$ it is the background logarithm in X_i. For each plaquette pair use one common colour coordinate for its readouts throughout the comparison. The held aligned pair at level j, modulo a simultaneous colour rotation, is the same measurable function of the retained state in both adjacent steps. One may choose any measurable representative; each box comparison may rotate both reads together because the polynomials used below are invariant. A simultaneous colour rotation preserves the scalar colour covariance in eq. 5.iv. Independent rotations of the two reads require the corresponding transport in that covariance. Individual curvature squares are gauge invariant.

Here are the inputs from the reference-window comparison. They concern the complete states of the original run on the reference torus. Write λ_j for the unsourced standing law with the block field and stored slots, and Π_j for the positive joint law of the original step from λ_{j-1} to λ_j. Its marginals are these two laws. Use bounded measurable extensions of the curvature reads to all states, with the same read in both adjacent comparisons. At the witness set the curvature coordinate to zero whenever its passed-test indicator fails. Its square then carries exactly the indicator of X_i. Require the extensions to obey

$$|Y_{j,p}| \le C g_x P$$

on every retained state, including every excluded part, with the common factor P fixed below. On the witness passed-test set this bound must hold for the actual read, so the extension does not change X_i. Outside admitted reads an extension can be clipped or set to zero. Boundedness without the factor g_x would not control the excluded mass relative to the signal.

At step j, the admitted part of Π_j is selected by a good event measurable in the held exterior, after summing the standing parts. The selector is constant on the product window; its complement has mass p_j. On the good exterior the normalized box law below is the disintegration of this original law. The same box integral normalizes the conditional measure and occurs in the exterior mixture weights. The read set contains every pivot and contour dependency of the pair. A restriction on an already deleted slot needs its own comparison and is not imposed by this selector. The

argument uses the original adjacent laws without conditioning the entire path on all good events simultaneously.

For each pair of plaquettes and each step let S be this finite set of free coordinates. The following bounds hold uniformly over the admitted exterior and all such pairs. A common factor $P = (\log x)^B \geq 1$ bounds the witness room, read dimension, maximum absolute free coordinate on the window, scaled curvature $g_x^{-1}|Y_{j,p}|$, and the polynomial chart constants used below. Increasing the fixed exponent B is allowed.

(1) At fixed exterior the normalized actual marginal on S has density ratio in $[e^{-\xi_j}, e^{\xi_j}]$ with respect to the same normalized flat box-window marginal. Here $\xi_j \leq 1$. Both flat Gaussians before restricting any window are centred in the same free coordinates. The local box Gaussian has window-tail probability at most $t_j \leq 1/2$. Its unwindowed covariance compressed to S differs in Hilbert–Schmidt norm by at most d_j from the corresponding full flat torus marginal. Both compressed covariances have eigenvalues in one fixed positive interval. This comparison includes the coarea and gauge densities, the full interaction, the exterior force and the cut. The marginal before removing the cut is on the same coordinates as the one used in the readout.

(2) On each selected full product window, the bounded previous-read extension agrees with a curvature readout $T_{j,p}(y; e)$, and the passed-test indicators in that extension equal one. Here y consists of the free coordinates on S and e is the held exterior. The retained read $Y_{j,p}$ is held there. The direct readout estimate is

$$\sup_{y \text{ in window}} |T_{j,p}(y; e) - Y_{j,p} - B_{j,p}^{\mathrm{loc}} y| \leq C g_x P^4 (\epsilon_j + g_x). \tag{5.iii}$$

The linear row $B_{j,p}^{\mathrm{loc}}$ is the flat row for this local read, using the actual physical weight of the step. Embed it by zero into the full flat coordinate set. There is a full flat row $B_{j,p}$ on that set with

$$\|B_{j,p}^{\mathrm{loc}} - B_{j,p}\|_{2\to 2} \leq C g_x P \epsilon_j, \qquad \|B_{j,p}\|_{2\to 2} \leq C g_x L^{-2(j-1-j_0)}.$$

Choose the full centred flat comparison law γ_j^0 to be the zero-next-datum soft-contour Gaussian of [3, Proposition 3.3], with its fixed positive contour weight. Its free-coordinate covariance is uniformly bounded on the admitted relative weight range. Write $\widehat{g}_{j-1} = \theta_{j-1}^{-1/2}$ and $\tau_{j-1} = x_{j-1}/\theta_{j-1}$. The full row $B_{j,p}$ is $u^{-2}\widehat{g}_{j-1}$ times its linear lifted-curvature row. The local comparison must use this same flat coordinate system and row. By [3, Propositions 3.3 and 4.3], scaling the action by τ_{j-1} while keeping the contour weight fixed scales its curvature covariance by τ_{j-1}^{-1}. Consequently

$$\mathbb{E}_{\gamma_j^0}[(B_{j,p} y) \otimes (B_{j,q} y)] = u^{-4} x_{j-1}^{-1} (P_{j-1} - P_j)(p, q)\, I_3. \tag{5.iv}$$

Here $\mathcal{C}_{(j)} = P_{j-1} - P_j$ is the unit raw-flux slice. The physical factor is exactly $\widehat{g}_{j-1}^2 \tau_{j-1}^{-1} = x_{j-1}^{-1}$.

(3) On the admitted terminal part, $|Y_{j_1,p}| \leq C g_x P L^{-2n}$. Its complement has mass p_{top}, and the bounded extension is used there. The choices of radii, localization buffers and excluded masses give

$$P^{50}\Delta \longrightarrow 0, \qquad \Delta := L^{-2n} + n g_x + p_{\mathrm{top}} + \sum_{j_0 < j \leq j_1} (p_j + \xi_j + \sqrt{t_j} + d_j + \epsilon_j). \tag{5.v}$$

The fixed power 50 is a sufficient reserve for the finite degree comparison below. The geometry and real window widths fix B before the analytic decay and window-tail reserves are required. For example, the excluded masses are superpolynomially small if their events are covered by at most a fixed power of x local contacts throughout the segment, each

of probability at most $Ce^{-c(\log x)^p}$, with $p > 1$. This union bound concerns the original adjacent laws and their actual determining sets.

One sufficient source of eq. 5.iii is a product window centred at zero, with $\|y\|_2 \le P^{3/2}$, and the estimates

$$|T_{j,p}(0;e) - Y_{j,p}| \le Cg_xP\epsilon_j,$$
$$\|D_yT_{j,p}(0;e) - B^{\mathrm{loc}}_{j,p}\| \le Cg_xP\epsilon_j,$$
$$\sup_{y \text{ in window}} \|D^2_yT_{j,p}(y;e)\| \le Cg^2_xP.$$

Taylor's formula along $[0,y]$ then bounds the constant, linear and quadratic errors by $Cg_xP\epsilon_j$, $Cg_xP^{5/2}\epsilon_j$ and $Cg^2_xP^4$, respectively. The direct readout estimate can instead be supplied without these derivative hypotheses.

The normalized density-ratio estimate also has a precise sufficient condition. At fixed exterior, suppose the actual conditional density on the same cut is proportional to e^{-V} times the flat reference probability, where V is real and its oscillation is at most ξ_j. Its normalizing integral lies between $e^{-\sup V}$ and $e^{-\inf V}$, so the normalized ratio lies in $[e^{-\xi_j}, e^{\xi_j}]$. Conditional expectation gives the same bounds for the marginal on S. Establishing this oscillation requires the complete conditional interaction and exterior force on that cut. A bound on only its quadratic part does not suffice.

Lemma 5.3 (Transport of polynomial readouts)**.** Under the adjacent-law inputs above, take two read plaquettes p, q and a homogeneous polynomial H of degree $m \in \{0, 2, 4\}$ in their six scaled curvature coordinates. Its sum of absolute coefficients is at most one, and it is invariant under simultaneous $SO(3)$ rotation of the two reads. Put $W_j = (Y_{j,p}, Y_{j,q})$ and let Z_j have the law of the six-dimensional vector of full flat readouts of γ^0_j, with the Z_j independent over levels. For all sufficiently large x, uniformly on the stated input family,

$$\left| \mathbb{E}_{\lambda_{j_0}} H(W_{j_0}) - \mathbb{E}H\left(\sum_{j_0 < j \le j_1} Z_j \right) \right| \le Cg^m_xP^{20}\Delta. \tag{5.vi}$$

A constant polynomial has zero error. ∎

Proof. First, at a given fixed exterior replace the actual read by $W_j + (B^{\mathrm{loc}}_{j,p}y, B^{\mathrm{loc}}_{j,q}y)$. The direct estimate eq. 5.iii bounds this replacement error by $Cg_xP^4(\epsilon_j + g_x)$. Increase the floor for x until every $\epsilon_j \le 1$, as permitted by eq. 5.v. The retained read and the linear Gaussian increments on that window have size at most Cg_xP^3, increasing fixed constants if necessary.

Define polynomials

$$H_j(w) = \mathbb{E}H\left(w + \sum_{j_0 < l \le j} Z_l \right), \qquad H_{j_0} = H.$$

Every slice has scalar colour covariance in the common pair frame. Its law, and therefore each H_j, is invariant under a simultaneous rotation of the two reads. Expansion of H into monomials bounds each derivative and moment norm by the sum of its individual bounds. The bound on the full rows and the geometric sum of their squared sizes give bounded moments of each coordinate of the Gaussian sum, of order $O(g^b_x)$ for a moment of degree b. This expansion bounds its derivative of order r at $|w| \le Cg_xP^4$ by $Cg^{m-r}_xP^{4(m-r)}$. This gives a local Taylor replacement error bounded by $Cg^m_xP^{20}(\epsilon_j + g_x)$ for H_{j-1}.

The density ratio in the first input changes a bounded window expectation by at most $C\xi_j$ times its supremum. To remove the window, write the full expectation as the two integrals on the window and its complement. Cauchy–Schwarz bounds the latter by $\sqrt{t_j}$ times the L^2 norm of the polynomial. The same Gaussian moment estimates, with the held W_j bounded by Cg_xP, bound this norm by $Cg^m_xP^{20}$. The normalization of the window costs at most a factor two. For

the difference of unwindowed compressed Gaussians interpolate their covariance matrices linearly. If C_v is the interpolant and D the covariance difference, the derivative of its density is that density times

$$\tfrac{1}{2}(y^{\mathsf{T}}C_v^{-1}DC_v^{-1}y - \operatorname{tr}(C_v^{-1}D)).$$

The second moment of the last factor is $\frac{1}{2}\|C_v^{-1/2}DC_v^{-1/2}\|_{\mathrm{HS}}^2$. Uniform bounds on the compressed covariances and Cauchy–Schwarz against the polynomial therefore give error $Cg_x^mP^{20}d_j$. On the full flat torus now use the same Gaussian to replace the local row by the full row. Its linear read difference has every fixed L^b norm at most $C_bg_xP\epsilon_j$. The polynomial identity for differences and Gaussian moments give the same cost with ϵ_j.

On the good exterior the estimates compare $H_{j-1}(W_{j-1})$ under the original box law with $H_j(W_j)$ at the held exterior. Their uniformity permits integration against the original exterior weights. On the excluded part, bounded extensions and Gaussian moments bound both quantities by $Cg_x^mP^{20}$, so its cost is at most $Cg_x^mP^{20}p_j$. The marginal identities give

$$\begin{aligned}&\mathbb{E}_{\lambda_{j_0}}H(W_{j_0}) - \mathbb{E}_{\lambda_{j_1}}H_{j_1}(W_{j_1})\\ &\qquad = \sum_{j_0<j\le j_1}\mathbb{E}_{\Pi_j}\big[H_{j-1}(W_{j-1}) - H_j(W_j)\big].\end{aligned}$$

A carried observable is the Radon–Nikodym derivative of its weighted pushforward with respect to the uninserted pushforward, as one checks against bounded functions of the output. At a merge both measures are sums with the original weights. Thus this identity preserves the exterior distribution. At the top, compare $H_{j_1}(W_{j_1})$ with $H_{j_1}(0)$ on the admitted part and use the bounded extension on its complement. The cost is $Cg_x^mP^{20}(L^{-2n}+p_{\mathrm{top}})$. Summing proves eq. 5.vi. Comparing full moments before forming their covariance retains the covariance of conditional means. □

Proposition 5.4 (The reference covariance)**.** Suppose the witness and clean-window inputs of lemmas 5.2 and 5.3 hold on one reference torus. Use the admitted flat hierarchy of lemma 3.4 through the two stops. Assume the enlarged supports are in its separated kernel range at the witness, and $x_{j-1} = x + O(n+1)$ on the segment. Then

$$S_{2;\star}^T(f_1,f_2) = \frac{3}{2}\zeta^2x^{-2}\sum_{p,q}\widetilde{f}_1(p)\widetilde{f}_2(q)G_{j_0}(p,q)^2 + o(x^{-2})M_1M_2. \tag{5.vii}$$

The error is uniform in the admitted bare data and positions. ■

Proof. Let $G_{j_0} = C_{K,s}^{\mathrm{bg}}$ and G_{j_1} be its nested coarser kernel. By lemma 3.4,

$$|G_{j_0}(p,q)| \le Ca^4d^{-4}, \qquad |G_{j_1}(p,q)| \le Cu^4L^{-4n}.$$

On the same enlarged supports each unit slice is bounded by

$$|\mathcal{C}_{(j)}(p,q)| \le Cu^4L^{-4(j-1-j_0)}.$$

With the bounded witness extension, the passed-test Wilson read is $\frac{1}{2}u^4|Y_{j_0,p}|^2 + O(u^8|Y_{j_0,p}|^4)$. This expansion holds on the witness law, with both sides zero when the test fails. No global clean selection is removed. There are at most $C\widehat{d}^4u^{-4}$ plaquettes in each read support, including its fixed enlargement. Since $|Y_{j_0,p}| \le Cg_xP$ and $\widehat{d} \le P$, the Wilson Taylor error is at most $Cg_x^4P^8M_i$ in the one-point read. Each quadratic read is bounded by $Cg_x^2P^6M_i$ on the entire witness law, using the bounded extension. The resulting covariance error is at most $C(g_x^6P^{14}+g_x^8P^{16})M_1M_2 = o(x^{-2})M_1M_2$. Apply lemma 5.3 to the square of one curvature read and the product of squares at two plaquettes. Use Δ from eq. 5.v. After smearing, the factors u^4 and u^8 cancel the plaquette counts. The degree-two mean errors are at most $Cg_x^2P^{24}\Delta M_i$, and the degree-four product error is at most $Cg_x^4P^{28}\Delta M_1M_2$. Each Gaussian mean is bounded by $Cg_x^2P^4M_i$, while the actual quadratic mean has the P^6 bound just established. Subtracting the products of means thus costs at most $Cg_x^4P^{30}\Delta M_1M_2$. The reserve $P^{50}\Delta \to 0$ pays each of these losses. The polynomials $|Y_p|^2$ and

$|Y_p|^2|Y_q|^2$ are invariant under simultaneous colour rotations. Their coefficient norms are 3 and 9; rescaling them to the unit coefficient ball only changes the fixed constants. Thus the argument uses degree-two and degree-four invariant polynomials on a pair of reads.

For the sum of independent slices, Wick's formula gives the covariance of two half-squares as $3/2$ times the square of the one-colour covariance. Thus its kernel is $\frac{3}{2}[\sum_j x_{j-1}^{-1}\mathcal{C}_{(j)}(p,q)]^2$. In particular the square contains the cross terms between different steps. They would be lost by summing only the one-step covariances at a zero retained field. Replacing the weights x_{j-1}^{-1} by x^{-1} has kernel error $O(x^{-2}(n+1)u^4)$ before squaring, by the geometric slice bound. The full square error after smearing is $O(x^{-3}(n+1)\widehat{d}^8)M_1M_2 = o(x^{-2})M_1M_2$. The slice sum now telescopes to $G_{j_0} - G_{j_1}$. The cross product with G_{j_1} and its square have total smeared cost at most

$$Cx^{-2}\big[(\widehat{d}L^{-n})^4 + (\widehat{d}L^{-n})^8\big]M_1M_2.$$

Here $n = \lceil 2\log_L \bar{x}_s\rceil$ and $\widehat{d}$ is polylogarithmic in x, so this is $o(x^{-2})M_1M_2$. Equation eq. 5.ii proves the result. If a local computation uses witness-lattice translates, the bare covariance and the free kernel have the same value on these translates. One may make all the replacements after averaging them; no finer translation invariance of the block fields is used. □

6. A common normalization with a finite reference torus

We keep the bare datum fixed while changing volume. Write $\ell_\star$ for the reference side and assume every compared torus side is an integer multiple of it. Choose an allowed board side $l_c = L^m$ with $\ell_\star/l_c$ even and with room for a unit ball at fixed positive distance from its faces. This side can be fixed at the reference coupling. The periodic array of sources below is obtained by the same cell reflections on all of these tori. For a physical test f use $\|f\|_\sharp = \max(\|f\|_\infty, c_M \operatorname{Lip} f)$, with the fixed positive constant c_M of the SI source norm.

Lemma 6.1 (Replication of a reflected source)**.** Let w be a complex source in the interior of one board cell, with no sourced plaquette crossing a cell face. Form the reflected array W_ℓ, conjugating the source under an odd number of cell reflections. Then the Wilson partition integrals with Haar probability measures satisfy

$$0 \le Z_\ell(W_\ell) \le Z_{\ell_\star}(W_{\ell_\star})^{(\ell/\ell_\star)^4}. \tag{6.i}$$

The unsourced bare inverse coupling is the same on every torus. ■

Proof. First lengthen one side, keeping the other three fixed. Use the group of spatial bond variables on a slice as the integration space, with product Haar probability. Between two slices at a cell face there is no source. If their spatial bonds are u, v, the temporal bonds between slices give, on integration, the kernel obtained from $\prod_b c(u_b(v_b^h)^{-1})$ by averaging the slice gauge h, where $c(g) = e^{-\beta(1-\operatorname{Tr} g/2)}$ and v_b^h is the usual site gauge action on the bond b. This is the crossing-plaquette product after cyclically permuting its traces. Thus the crossing operator is $K = CP$, where C is convolution by the product of c over slice bonds and P is Haar averaging over slice gauges. Centrality of c makes the convolution commute with the left and right group actions on each bond, hence with P.

The expansion $c = e^{-\beta}\sum_{n\ge0}(\beta/2)^n\chi_{1/2}^n/n!$ has nonnegative character coefficients. Indeed the powers are characters of tensor powers of the defining representation. At the identity the sum of coefficients times the representation dimensions is one. For one bond a coefficient a_λ in dimension d_λ gives convolution eigenvalue a_λ/d_λ with multiplicity d_λ^2. Consequently C is positive trace class on the finite product slice space. The commuting orthogonal projection P makes K positive trace class as well. Its square root S is Hilbert–Schmidt.

Let B be the transfer through one slab between two board faces with the crossing factors at its ends removed. It is the product of the spatial multiplication factors and the integrated interior temporal kernels. These are bounded operators since the slice space is a compact product with

finite Haar measure and the profile is bounded. The next slab is the reflected conjugate slab and has operator B^*. This follows by reversing the order of the slices and of the temporal bond orientations, which preserves Haar measure and the plaquette traces. Any transverse board array is held fixed in this argument. With $A = SBS$, the partition integral over a whole period is $\operatorname{Tr}(AA^*)^h$ if the period has $2h$ slabs. Indeed before splitting each crossing factor it is the cyclic integral of $KB, KB^*, \ldots$. Each of these factors has a Hilbert–Schmidt kernel. The trace identity for a product of two Hilbert–Schmidt kernels, with the other factors composed by Fubini, gives that cyclic integral. Cyclicity of the trace then splits the crossing factors into the two square roots. This also covers a slab consisting of a single time slice, when B is a multiplication operator.

The positive trace-class operator $T = (AA^*)^h$ has $\operatorname{Tr} T^b \le (\operatorname{Tr} T)^b$ for every positive integer b, by its nonnegative eigenvalues. Repeating the period b times gives exactly the array on the longer side. This proves the one-direction inequality. The number of cells is even in every direction, so successive replication in all four directions preserves the same array. The intermediate rectangular tori use only finite Wilson integrals. They require no effective-action construction. This proves eq. 6.i. □

Lemma 6.2 (Comparison of vacuum densities)**.** Let $b_j = L^{j-K}$, $0 \le j \le K$. Suppose the once-counted unsourced vacuum on each compared torus has the form

$$E_{\ell,K} = \sum_{j=0}^{K} \left[(\ell/b_j)^4 c_{j,K} + \sum_{z \in (b_j\mathbb{Z}/\ell\mathbb{Z})^4} \sum_{D \ni z} v_{\ell,j}(D, z) \right].$$

The numerical coefficients $c_{j,K}$ and the fixed auxiliary data are common. There are rooted infinite companions $v_{\infty,j}$ with the same translation-covariant sums per root. Suppose their rooted domain counts at length m are bounded by $Ce^{\kappa_D m}$, and their individual absolute values by $Ce^{-\kappa_E m}$ with $\kappa_E > \kappa_D$. The same bounds hold for the finite domains. At each root assume a bijection of finite and infinite rooted reads of length $m < \ell/(C_0 b_j)$ which preserves their vacuum values. Then for $\ell \ge \ell_0 > 0$ there is a common number A_K such that

$$\left| \ell^{-4} E_{\ell,K} - A_K \right| \le C \sum_{j=0}^{K} b_j^{-4} e^{-c\ell/b_j} \le C_{\ell_0} e^{-c\ell/2}. \tag{6.ii}$$

The constants are independent of the cutoff. The number A_K is allowed to depend on the cutoff and the common bare datum. ■

Proof. The companion sum $\bar{v}_j$ at one root is absolutely convergent. Set $A_K = \sum_j b_j^{-4}(c_{j,K} + \bar{v}_j)$. Short rooted values cancel in the difference. The two tails are bounded by geometric sums with the positive gap $\kappa_E - \kappa_D$. Dividing by the physical volume yields the first inequality. For the second, put $u = K - j$ and bound

$$\sum_{u \ge 0} L^{4u} e^{-c\ell L^u} \le e^{-c\ell/2} \sum_{u \ge 0} L^{4u} e^{-c\ell_0 L^u/2}.$$

The last sum converges. Thus A_K cancels exactly in a comparison with the reference torus and need not converge. In the scalar convention of [4, Proposition 2.5] the numerical free-bond and gauge factors go in $c_{j,K}$. One can point an unpointed remainder by allocating its value equally to its roots. The assumed equality is an equality of the actual rooted vacuum values on nonwrapping complete reads. Equality of bare weights alone does not imply it. □

Proposition 6.3 (One analytic cell subtraction for both measures)**.** Use the comparison geometry and the hypotheses of lemma 6.2. Suppose the actual reference torus has the ordinary complex source upper bound of [5, Proposition 8.1]

$$|Z_{\ell_\star}(W)| \le \exp\{E_{\ell_\star,K} + \operatorname{Re}\mathcal{E}_\star(W) + C_+\ell_\star^4\},$$

and every compared torus, including the reference, has the unsourced floor $Z_\ell(0) \ge \exp\{E_{\ell,K} - C_-\ell^4\}$. Assume the constants are uniform at the fixed reference coupling. In the reference source number, each domain term is holomorphic in the source on the ordinary radius $2r$, vanishes at zero and depends only on its complete source read. At scale b_j root each term at a level site in its complete read. At length m assume the absolute bound $Ce^{-\kappa_E m}$ on the ordinary source ball. Require the complete read together with its root to have diameter at most $Cb_j(m+1)$ and the rooted count to be at most $Ce^{\kappa_D m}$, with $\kappa_E > \kappa_D$. Separate point terms, if used, are holomorphic on the same ordinary source ball, vanish at zero and have only their own point read.

For any real Lipschitz tests F_1, F_2 supported in one unit ball, there is one holomorphic function $E^\circ(z)$, $E^\circ(0) = 0$, for which

$$\left|e^{-E^\circ(z)}\omega(e^{z_1\mathcal{O}_a(F_1)+z_2\mathcal{O}_a(F_2)})\right| \le Q \tag{6.iii}$$

on $\sum_i |z_i|\|F_i\|_\sharp < r$. Here ω can be the reference measure, any compared torus measure or any periodic Wilson Gibbs state at this datum. For the last assertion assume the compared tori contain a compatible cofinal sequence. The constants are uniform in the cutoff, volume, position and states. ∎

Proof. Place the common support ball inside the chosen cell by a joint bare lattice shift. Translation invariance lets us return to the original position. Use the cell source $w_z = z_1F_1 + z_2F_2$ and its reflected conjugate array W_ℓ. The even and odd profiles separately vanish near faces, so joining copies does not increase their Lipschitz constants. Using the two sets of coefficients z_i and $\bar{z}_i$ gives ordinary source size at most $2\sum_i |z_i|\|F_i\|_\sharp$. The reflected dissemination of $e^{\mathcal{O}_a(w_z)}$ is $e^{\mathcal{O}_a(W_\ell)}$, since each sourced plaquette is internal and reflection preserves its trace. With $N_\ell = (\ell/l_c)^4$, its finite chessboard seminorm is $[Z_\ell(W_\ell)/Z_\ell(0)]^{1/N_\ell}$. The numerator is nonnegative. By lemma 6.1 and the two partition bounds, this seminorm is at most

$$\exp\left\{N_{\ell_\star}^{-1}\operatorname{Re}\mathcal{E}_\star(W_{\ell_\star}) + Cl_c^4\right\}.$$

Here we used eq. 6.ii for the difference of the two vacuum densities. The common divergent number has cancelled.

Define E° by keeping the reference source terms whose complete read lies inside one open board cell and averaging over the cells. Keep every nonzero point term. Use the value of each even term and the conjugate of each odd term. The odd actual value is $h(\bar{z})$ for a holomorphic h, so $\overline{h(\bar{z})}$ is holomorphic and has the same real part. All the kept terms are evaluated on the source in their own cell. They vanish at zero. Normal convergence on smaller source discs follows from the rooted bounds at the fixed cutoff. This defines the same E° for all the measures in the statement.

A discarded nonzero term meets a source support and reads outside that support's cell. Its diameter is at least the fixed clearance. At scale b_j its length is at least $c/b_j - C$. Charge the term to one cell in which it reads a nonzero source. Its root is within $Cb_j(m+1)$ of the support ball in that cell, so there are at most $C(1+b_j^{-1})^4(m+1)^4$ possible roots. The strict entropy margin absorbs the polynomial in m. Summing over the allowed lengths and then the scales, the omitted absolute source number per cell is bounded by

$$C\sum_{u\ge 0}(1+L^u)^4e^{-c'L^u}, \qquad u = K - j.$$

The constant incorporates the fixed localization side and collar. Thus the finite seminorm after subtracting E° is bounded by a common Q. The one-cell chessboard inequality gives eq. 6.iii on

the tori. The fixed-spacing all-state board bound of [1, Theorem C] gives it along the same cofinal family for every periodic state. This argument uses bounded cylinder exponentials at the fixed cutoff. In particular it does not select the state as a weak limit of these particular tori. □

Lemma 6.4 (Transfer of a local covariance)**.** Let two translation-invariant laws at the same bare spacing, with the same linear source convention, obey eq. 6.iii for every unit-ball test pair, with a common E° for the two laws. The tests in this assumption may overlap and have either sign. For a pair in definition 3.1, with $a \le d$ sufficiently small, the difference of their connected covariances satisfies

$$|S^T_{2;1}(f_1, f_2) - S^T_{2;2}(f_1, f_2)| \le C_\vartheta d^2 M_1 M_2. \tag{6.iv}$$

For tori, take supports and the shifted unit ball in one common lift. ■

Proof. For any unit-ball pair, apply the local residual Cauchy bound of [5] to e^{-E° times the two generating functions. Their values at zero are one. Equivalently the bound Q and radial Schwarz give a logarithm bounded by $\log 2$ on a bidisc whose coordinate radii are $r/[4(Q+1)\|F_i\|_\sharp]$. The two mixed derivatives of E° cancel. If D is the difference of the two covariances, then $|D(F_1, F_2)| \le C\|F_1\|_\sharp \|F_2\|_\sharp$. A zero test gives zero directly.

The two supports lie in one radius-$5d$ ball. Pack $J \ge cd^{-4}$ disjoint lattice translates of that ball inside a unit ball, using shifts on a grid with a lattice spacing comparable to d. For independent signs ε_v indexed by the shifts put $F_i = \sum_v \varepsilon_v \tau_v f_i$. Both the supremum and Lipschitz norms are unchanged or decreased, because the supports of the copies are disjoint and each test vanishes at its ball boundary. For points in different balls split their connecting segment at the two boundaries to obtain the Lipschitz assertion. The sign average of $D(F_1, F_2)$ is $JD(f_1, f_2)$ by bilinearity and joint translation invariance. Divide by J and use $\|f_i\|_\sharp \le C_\vartheta d^{-1} M_i$. This gives eq. 6.iv. Separate analytic normalizers would leave their mixed derivative difference in this calculation. □

Theorem A (The short-distance two-point law)**.** Use the witness and running source of propositions 4.1 and 4.3 at one fixed sufficiently small reference coupling, the actual reference comparison of proposition 5.4, and the common finite-reference inputs of proposition 6.3. Fix the common integer $n_0 \ge 1$ above which those formation and reference inputs are supplied, and require $K \ge s(d) + n_0$. Let (f, h) belong to Definition 3.1 at separation d. Then below a common $d_0 > 0$, on every admitted torus and in every admitted periodic Gibbs state,

$$S^T_{2;a,\rho}(f, h) = \frac{9}{\pi^4}\, \zeta_K(d)^2 g_K(d)^4 \mathcal{K}(f, h)(1 + r_{K,\rho}(f, h)), \qquad |r_{K,\rho}(f, h)| \le \tfrac{1}{2}. \tag{6.v}$$

Here ρ denotes the compared finite-volume law or periodic state; the remainder may depend on it. The error tends to zero as $d \downarrow 0$, uniformly over the bare data, support positions and compared measures for which the displayed input estimates are uniform. The two factors $\zeta_K(d)$ and $g_K(d)$ are those of the same run at the same bare datum throughout the comparison. ■□

Proof. To put the same pair on the reference torus, lift it to the common radius-$5d$ ball and extend each test by zero outside its supporting ball. Stopping a line segment at its first boundary point, where the test vanishes, preserves the Lipschitz bound. Periodization into any torus of side at least $40d$ preserves that bound, the masses, support gap and $\mathcal{K}$; split shortest segments at boundaries of the disjoint lifted copies. For the actual signed profiles $\widetilde{f}_i$, the tent estimates give $\|\widetilde{f}_i - f_i\|_\infty = o(1)M_i$ and an enlargement by at most CMa_s. The supports remain separated by at least $d/2$ below a common threshold, by the witness room. For small enough d, the witness bounds $a \le a_s$ and $d/a_s \to \infty$ give $10a \le d/2$, $a_s \le d/2$ and $d \le \ell_\star/200$. The enlarged supports then stay in the separated kernel range. There are at least two sites in each direction at the used coarse levels. Apply proposition 5.4 with these signed profiles. The separated free background bound $|G_{j_0}(p, q)| \le Ca^4 d^{-4}$ and the two plaquette counts show that replacing the signed profiles by the originals costs $o(x^{-2})M_1 M_2$. This uses an absolute estimate before any division by a positive main

term. The hierarchy and distance hypotheses of proposition 3.3 hold for the original pair. That proposition replaces the unit raw-flux square sum by $6\pi^{-4}\mathcal{K}(f,h)(1+o(1))$. The factor $3/2$ in eq. 5.vii then gives $9\pi^{-4}$.

The shape estimate and running-source lower bound give

$$\frac{9}{\pi^4}\zeta_K(d)^2x^{-2}\mathcal{K}(f,h)\geq c_\vartheta x^{-2}\|f\|_\infty\|h\|_\infty.$$

The relative reference error therefore tends to zero. The free replacement also gives

$$\sum_{p,q}f(x_p)h(x_q)G_{j_0}(p,q)^2\geq c_\vartheta M_f M_h.$$

Thus the reference covariance equals this unmodified lattice sum times $3\zeta^2x^{-2}/2$ with relative error at most $1/4$, after decreasing the common threshold. Apply lemma 6.4 with the finite reference and the given measure. Its relative error is $C_\vartheta d^2x^2$, which tends uniformly to zero since $x\asymp\log(1/d)$. The common-normalizer argument uses the same physical source with the same bare datum, without changing the Wilson coefficient or rescaling the insertion. Decrease d_0 until the total relative error is at most $1/2$. This proves eq. 6.v. □

Corollary 6.5 (Transfer with a cofinal reference)**.** For each fixed bare datum suppose there is one compatible cofinal sequence of finite tori on which eq. 5.vii holds for $K\geq s(d)+n_0$, with one fixed $n_0\geq 1$ and a common positive distance threshold for this cofinal comparison. Bound its additive remainder by $\epsilon_{\mathrm{ref}}(d)x^{-2}M_1M_2$, where one $\epsilon_{\mathrm{ref}}(d)\to 0$ works uniformly over the cofinal sides above a common positive minimum, all admitted bare data, shapes and support positions. Use the same actual profiles and hierarchy as in proposition 5.4. The sequence at one datum works for every fixed admissible pair there with the same witness and coefficient factors. Suppose the fixed-spacing state-transfer hypotheses of [5, Theorem A] hold uniformly, with a common cell subtraction for all periodic states at the datum. Then the conclusion of theorem A holds for the periodic states. It needs no comparison of vacuum densities with one fixed finite torus in this case. ■□

Proof. First replace the pointed profiles by the originals using their absolute tent error and the separated kernel bound. Apply proposition 3.3 on each torus; its error is uniform once the side is above the common minimum. Dividing the resulting additive error by the positive shape floor gives the same relative reference law as in the finite-reference proof. At fixed spacing take a local weak limit of the cofinal torus laws, using compactness and the finite-range Gibbs specification as in [1]. This gives a periodic Wilson Gibbs state at the same bare datum. The covariance of the fixed compact tests is built from bounded continuous cylinder functions at that spacing. The uniform reference estimate therefore passes to this state, with the same $\zeta_K(d)$ and $g_K(d)$. The common cell subtraction of [5, Theorem A] serves the limit state and any periodic state at that datum. Apply lemma 6.4 and the positive shape lower bound exactly as in the finite-reference proof. Uniformity over the cofinal sequence for fixed pairs is used before taking the volume limit. A distance threshold tending to zero along that sequence would not suffice for this passage. □

7. Ultraviolet bounds on the prescribed class

Corollary 7.1 (Bounds and conditional passage to a limit)**.** Under the hypotheses of Theorem A or Corollary 6.5, there are $c, C > 0$ such that, for all sufficiently small d and all admitted cutoffs, states and pairs,

$$\frac{c\vartheta^2 M_f M_h}{[\log(1/d)]^2} \le S^T_{2;a,\rho}(f,h) \le \frac{C}{[\log(1/d)]^2} \min\{M_f M_h, d^{-8} m_f m_h\}. \tag{7.i}$$

For each fixed separated pair, any limit of the connected correlations along admitted cutoffs obeys the same bounds. If such limits exist for every pair in the Euclidean class and $\vartheta \le \vartheta_*$, the limit satisfies the ultraviolet bound on $\mathfrak{T}_{\mathbb{R}^4}(d;\vartheta_*)$ with $u(d) = C[\log(1/d)]^{-2}$. For the mass-one dilates constructed below and $\vartheta \le \vartheta_*$, the connected correlations, and any corresponding fixed-pair limits, are bounded above and below by positive constant multiples of $d^{-8}[\log(1/d)]^{-2}$. On a family where the theorem gives uniform vanishing of the relative error as $d \downarrow 0$, the corresponding limiting correlations have leading coefficient C_G/c_*^2 multiplying $\mathcal{K}(f,h)[\log(1/d)]^{-2}$. ■□

Proof. At the witness, $\zeta_K(d) \to 1$ and $g_K(d)^{-2}/\log(1/d) \to c_* > 0$ uniformly in the admitted cutoff data. Shrink the distance threshold until the source factor and the inverse-coupling ratio have positive lower and finite upper bounds. The relative error bound $|r| \le 1/2$ puts $1 + r$ in $[1/2, 3/2]$. Use the two-sided integral estimate eq. 3.ii in the comparison formula of Theorem A. This proves eq. 7.i. Its constants are uniform along the admitted cutoff sequence, so the inequalities hold in any indicated limit. The inclusion at $\vartheta \le \vartheta_*$ follows from eq. 3.i.

For an explicit pair at scale one, take $f(x) = (12/\pi^2)(1 - |x|^2)_+^2$ and $h(x) = f(x - 4e_1)$. The radial integral of $(1-|x|^2)_+^2$ is $\pi^2/12$, and its largest radial derivative has magnitude $8/(3\sqrt{3})$. The support gap is two. These facts give $(f,h) \in \mathfrak{T}_{\mathbb{R}^4}(1;\vartheta_*)$. For a fixed x_0, use $f_d(x) = d^{-4} f((x-x_0)/d)$ and $h_d(x) = d^{-4} h((x-x_0)/d)$. Both masses stay one, and the amplitude product is a fixed positive constant times d^{-8}. Dilation and positive amplitude scaling preserve the shape conditions at scale d. For these tests eq. 7.i gives the claimed positive lower and upper bounds.

For the last assertion, divide the comparison by $C_G\mathcal{K}(f,h)/[c_*^2 \log^2(1/d)]$. The resulting three factors tend uniformly to one in the order stated, and every fixed-pair cutoff limit inherits the same uniform small-distance error. □

7.1. **Explicit tests at the fixed shape parameter.** The following check gives the explicit tests of the continuum application. Let A_0 be one on $[9/4, 11/4]$, zero outside $[17/8, 23/8]$, and use the ramp $3u^2 - 2u^3$, $0 \le u \le 1$, on each transition interval, with u increasing towards the plateau. Then $\int A_0 = 5/8$ and $\|A_0'\|_\infty \le 12$, by the integral and maximum derivative of the ramp. For $I = [-1/4, 1/4]$ put $B_0 = 12\mathbf{1}_I^{*4}$. Direct integration of the triangular convolution $\mathbf{1}_I^{*2}$ gives

$$B_0(s) = \begin{cases} 1 - 6s^2 + 6|s|^3, & |s| \le 1/2, \\ 2(1-|s|)^3, & 1/2 \le |s| \le 1, \\ 0, & |s| \ge 1. \end{cases}$$

It is C^2, $\|B_0\|_\infty = 1$, and $\|B_0'\|_\infty = 2$, with the maximum derivative attained at $|s| = 1/3$. The Fourier transform is $\widehat{B}_0(q) = \frac{3}{4}(\sin(q/4)/(q/4))^4$, with value $3/4$ at zero, by taking the fourth power of the transform of $\mathbf{1}_I$.

For transverse coordinates $z \in \mathbb{R}^2$, take $q(s) = B_0(4s/3)$, $D(s) = \sum_{j\in\mathbb{Z}} q(s-j)$ and $\chi = q/D$. The nearest integer yields $D \ge 2/27$, and at most two summands are nonzero at each point. Since $q \le D$, the quotient rule gives $\|\chi'\|_\infty \le 108$. The integer translates of $\psi(z) = \chi(z_1)\chi(z_2)$ sum to one by periodicity of D. Let $\widetilde{\psi}(z) = \chi_0(z_1)\chi_0(z_2)$, where $\chi_0 = 1$ on $[-1,1]$, has support $[-3/2, 3/2]$ and uses the same ramp. The support of ψ lies in $[-3/4, 3/4]^2$, so $0 \le \psi \le \widetilde{\psi}$. The product rule gives $\|\nabla\psi\|_\infty \le 216$ and $\|\nabla\widetilde{\psi}\|_\infty \le 6$. Integration gives $\int \widetilde{\psi} = (5/2)^2$.

For a nonzero real $\varphi \in C_c^1(\mathbb{R}^2)$, $\delta > 0$, $1 \le \lambda \le 2$ and $i \in \mathbb{Z}^2$, set

$$A = \|\varphi\|_\infty + \delta\|\nabla\varphi\|_\infty, \qquad \psi_i(z) = \psi(z/\delta - i), \qquad \widetilde{\psi}_i(z) = \widetilde{\psi}(z/\delta - i).$$

Put $\varphi_i^- = 2A\widetilde{\psi}_i$ and $\varphi_i^+ = 2A\widetilde{\psi}_i + \psi_i\varphi$. The two functions

$$H_i^\pm(t,x,z) = A_0(t/\delta)B_0(4x/(\lambda\delta))\varphi_i^\pm(z)$$

lie between A and $3A$ times the product of the two one-dimensional factors with $\widetilde{\psi}_i$. In particular their sup norms lie in $[A, 3A]$. The sum of $H_i^+ - H_i^-$ over the finitely many indices with $\psi_i\varphi \neq 0$ equals $A_0(t/\delta)B_0(4x/(\lambda\delta))\varphi(z)$. Their supports lie in balls of radius $d = 5\delta/2$, since the sum of squared coordinate half-widths around $(5\delta/2, 0, \delta i)$ is at most $5\delta^2$. Each function and its time reflection have separation between 4δ and 6δ, by the time support and the reflected displacement of a point on the plateau. The mass is at least $375A\delta^4/512$ by integrating the product lower bound and using $\lambda \ge 1$. The Lipschitz constant is bounded by $A[3(12+8) + 2\cdot 6 + 216 + 1]/\delta$. Dividing the mass by $3A(5\delta/2)^4$ gives $1/160$. The derivative bound is at most $12000A/d$. Thus $(H_i^\pm \circ \theta, H_i^\pm)$ belongs to $\mathfrak{T}_{\mathbb{R}^4}(5\delta/2; \vartheta_*)$, where $\theta(t,x,z) = (-t,x,z)$.

References

[1] *Periodic Wilson states and reflection positivity*, companion manuscript, 2026.

[2] *Linear lattice PDE with block constraints*, companion manuscript, 2026.

[3] *Gaussian block integration and free action-density correlations*, companion manuscript, 2026.

[4] *Coupling trajectories and partition stability in lattice Yang–Mills theory*, companion manuscript, 2026.

[5] *Scalar sources and uniform insertions for lattice Yang–Mills fields*, companion manuscript, 2026.

[6] T. Bałaban, *Renormalization group approach to lattice gauge field theories. I. Generation of effective actions in a small field approximation and a coupling constant renormalization in four dimensions*, Comm. Math. Phys. **109** (1987), 249–301. doi:10.1007/BF01215223.